Unit 2: First Order Linear Differential Equations

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Master Unit Reference Guide: Cheat Sheet, Problems, and Solutions

Contents

Unit 2 at a Glance: Core Sub-Topics

This unit assembles the core solving toolkit of first order theory. Each sub-topic below builds on the last, moving from raw integration techniques toward genuine physical models.

- **Separable Equations** — splitting a rate law into a pure x side and a pure y side, then integrating each independently.
- **First-Order Linear Equations and Integrating Factors** — the standard form $y' + P(x)y = Q(x)$ and the factor that collapses the left side into a single derivative.
- **Initial Value Problems** — using a known data point to pin the arbitrary constant and select one curve from the solution family.
- **Homogeneous Equations and Substitution** — the substitution $y = vx$ that reduces a degree-matched equation to a separable one.
- **Applications: Growth, Decay, and Newton's Law of Cooling** — the exponential models that these methods unlock.

Part I — Condensed Cheat Sheet

2.1 Separable Equations

A first order equation is *separable* when its right side factors into a function of x alone times a function of y alone — that single structural test is the whole game. Once it factors, you divide and multiply until every y rides with dy and every x rides with dx, then integrate both sides.

Standard Form and Method: Separable

A separable equation and its solution scheme:

$$\frac{dy}{dx} = g(x)\,h(y) \quad\Longrightarrow\quad \int \frac{1}{h(y)}\,dy = \int g(x)\,dx + C.$$

One constant C suffices, since the two constants of integration merge into a single arbitrary constant.

Common Pitfall: Lost Equilibrium Solutions

Dividing by $h(y)$ silently assumes $h(y) \neq 0$. Any constant value of y that makes $h(y) = 0$ is an equilibrium solution that this step can quietly discard — always check it separately.

Worked Example 2.1: Solve $\frac{dy}{dx} = 6x^2 y$

Step 1. Separate the variables:

$$\frac{1}{y}\,dy = 6x^2\,dx.$$

Step 2. Integrate both sides:

$$\ln|y| = 2x^3 + C.$$

Step 3. Exponentiate to isolate y:

$$y = e^{2x^3 + C} = C\,e^{2x^3}.$$

The dropped solution $y = 0$ is recovered as the case $C = 0$.

2.2 First-Order Linear Equations and Integrating Factors

A first order equation is *linear* when y and y' each appear only to the first power and never multiply one another. Before reading off anything, you must normalize so the derivative carries coefficient one — only then is $P(x)$ truly the coefficient of y.

Standard Form and Integrating Factor

Standard form, integrating factor, and solution:

$$\frac{dy}{dx} + P(x)\,y = Q(x), \qquad \mu(x) = e^{\int P(x)\,dx},$$

$$\frac{d}{dx}\left[\mu(x)\,y\right] = \mu(x)\,Q(x) \quad\Longrightarrow\quad y = \frac{1}{\mu(x)}\left(\int \mu(x)\,Q(x)\,dx + C\right).$$

Algorithm: Solving a First-Order Linear Equation

1. Write the equation in standard form $y' + P(x)\,y = Q(x)$, dividing through so the derivative carries coefficient one.
2. Compute the integrating factor $\mu(x) = e^{\int P(x)\,dx}$.
3. Multiply through by $\mu(x)$; the left side collapses to the single derivative $\frac{d}{dx}\big[\mu(x)\,y\big]$.
4. Integrate both sides, then divide by $\mu(x)$ to isolate y.

The magic of $\mu(x)$ is that multiplying through turns the entire left side into the exact product-rule derivative $\frac{d}{dx}[\mu y]$ — that is the single fact the whole method rests on.

Worked Example 2.2: Solve $\frac{dy}{dx} + y = e^x$

Step 1. Read off $P(x) = 1$, so the integrating factor is

$$\mu = e^{\int 1\,dx} = e^x.$$

Step 2. Multiply through; the left side collapses to a derivative:

$$\frac{d}{dx}\big[e^x y\big] = e^x \cdot e^x = e^{2x}.$$

Step 3. Integrate both sides:

$$e^x y = \frac{1}{2}e^{2x} + C.$$

Step 4. Divide by e^x:

$$y = \frac{1}{2}e^x + Ce^{-x}.$$

2.3 Initial Value Problems

The general solution of a first order equation is a whole family of curves indexed by one arbitrary constant. An *initial condition* $y(x_0) = y_0$ selects exactly one member — it fixes the constant to the value that forces the curve through the given point.

General versus Particular Solution

A first order equation needs exactly **one** initial condition. The general solution carries the free constant; the particular solution has it determined:

$$y = F(x) + C \quad \xrightarrow{y(x_0)=y_0} \quad C = y_0 - F(x_0).$$

Worked Example 2.3: Solve $\frac{dy}{dx} = 3x^2$, $y(1) = 5$

Step 1. Integrate for the general solution:

$$y = x^3 + C.$$

Step 2. Apply $y(1) = 5$:

$$5 = (1)^3 + C \implies C = 4.$$

Step 3. State the particular solution:

$$y = x^3 + 4.$$

2.4 Homogeneous First-Order Equations and Substitution

A function $f(x, y)$ is *homogeneous of degree* n when scaling both inputs pulls out a clean power of the scale factor: $f(tx, ty) = t^n f(x, y)$. An equation $M\,dx + N\,dy = 0$ is homogeneous when M and N share the same degree — and that shared degree is precisely what lets the ratio substitution work.

The Substitution $y = vx$

For a homogeneous equation, set

$$y = vx, \qquad \frac{dy}{dx} = v + x\frac{dv}{dx},$$

which always reduces the equation to a *separable* one in v and x. After solving for v, back-substitute $v = \dfrac{y}{x}$.

Worked Example 2.4: Solve $\frac{dy}{dx} = \frac{x+2y}{x}$

Step 1. Rewrite the right side and substitute $y = vx$:

$$\frac{dy}{dx} = 1 + 2\frac{y}{x} = 1 + 2v, \qquad v + x\frac{dv}{dx} = 1 + 2v.$$

Step 2. Cancel v and separate:

$$x\frac{dv}{dx} = 1 + v \implies \frac{1}{1+v}\,dv = \frac{1}{x}\,dx.$$

Step 3. Integrate and exponentiate:

$$\ln|1 + v| = \ln|x| + C \implies 1 + v = Cx.$$

Step 4. Back-substitute $v = \frac{y}{x}$ and clear:

$$1 + \frac{y}{x} = Cx \implies y = Cx^2 - x.$$

2.5 Applications: Growth, Decay, and Newton's Law of Cooling

When a rate of change is proportional to the current amount, the solution is exponential — this single idea drives population models, radioactive decay, and cooling alike.

Exponential Models

Growth and decay:

$$\frac{dP}{dt} = kP \implies P(t) = P_0 e^{kt} \qquad (k > 0 \text{ grows},\ k < 0 \text{ decays}),$$

with doubling time $t = \dfrac{\ln 2}{k}$. Newton's law of cooling:

$$\frac{dT}{dt} = -k\,(T - T_s) \implies T(t) = T_s + (T_0 - T_s)\,e^{-kt},$$

so $T \to T_s$ as $t \to \infty$.

Worked Example 2.5: A culture doubles in 3 hours

Step 1. The model $P = P_0 e^{kt}$ doubles when $e^{3k} = 2$:

$$3k = \ln 2 \implies k = \frac{\ln 2}{3}.$$

Step 2. The growth law is therefore

$$P(t) = P_0\, e^{(\ln 2/3)\,t} = P_0\, 2^{\,t/3}.$$

Every 3 hours the population multiplies by two, as required.

Part II — Review Guide: Practice Problems

Work the following ten problems in order; they are graded from direct separation through linear factors, initial value problems, the homogeneous substitution, and a full physical model. Solutions follow in Part III.

1. Solve $\dfrac{dy}{dx} = \dfrac{x}{y}$ by separation of variables.

2. Solve $\dfrac{dy}{dx} = e^{x-y}$.

3. Solve $\dfrac{dy}{dx} = y\cos x$, and identify any solution that is lost when you divide by y.

4. Solve the linear equation $\dfrac{dy}{dx} + 3y = 6$ using an integrating factor.

5. Solve $x\dfrac{dy}{dx} + y = x^2$ by first writing it in standard linear form.

6. Solve $\dfrac{dy}{dx} + \dfrac{2}{x}y = x$.

7. Solve the initial value problem $\dfrac{dy}{dx} + 2y = 4$ with $y(0) = 1$.

8. Solve the homogeneous equation $\dfrac{dy}{dx} = \dfrac{x+y}{x}$ using the substitution $y = vx$.

9. Solve the homogeneous equation $\dfrac{dy}{dx} = \dfrac{x^2+y^2}{xy}$ using $y = vx$.

10. **(Application)** An object at 100° is placed in a room held at 20°. After 10 minutes its temperature is 60°. Find the temperature function $T(t)$ and determine how long it takes the object to reach 30°.

Part III — Complete Solutions

Solution 1. Separate, then integrate.

$$y\,dy = x\,dx$$

$$\frac{y^2}{2} = \frac{x^2}{2} + C_1$$

$$y^2 = x^2 + C \qquad \text{(general solution).}$$

Solution 2. Split the exponential as $e^{x-y} = e^x e^{-y}$, then separate.

$$e^{\,y}\,dy = e^{\,x}\,dx$$

$$e^{\,y} = e^{\,x} + C$$

$$y = \ln(e^{\,x} + C).$$

Solution 3. Separate (assuming $y \neq 0$), integrate, and exponentiate.

$$\frac{1}{y}\,dy = \cos x\,dx$$

$$\ln|y| = \sin x + C$$

$$y = C\,e^{\sin x}.$$

Dividing by y assumed $y \neq 0$, so it dropped the equilibrium solution $y = 0$ — which is recovered as the special case $C = 0$.

Solution 4. Here $P = 3$, so the integrating factor is $\mu = e^{\int 3\,dx} = e^{3x}$. Multiply through:

$$\frac{d}{dx}\left[e^{3x}y\right] = 6\,e^{3x}$$

$$e^{3x}y = \int 6\,e^{3x}\,dx = 2\,e^{3x} + C$$

$$y = 2 + C\,e^{-3x}.$$

Solution 5. Divide by x to reach standard form:

$$\frac{dy}{dx} + \frac{1}{x}\,y = x, \qquad \mu = e^{\int (1/x)\,dx} = e^{\ln|x|} = x.$$

Multiply through and integrate:

$$\frac{d}{dx}\left[xy\right] = x \cdot x = x^2$$

$$xy = \frac{x^3}{3} + C$$

$$y = \frac{x^2}{3} + \frac{C}{x}.$$

Solution 6. With $P = \frac{2}{x}$, integrate the coefficient and exponentiate:

$$\mu = e^{\int (2/x)\,dx} = e^{2\ln|x|} = x^2.$$

Multiply through and integrate:

$$\frac{d}{dx}\left[x^2 y\right] = x^2 \cdot x = x^3$$

$$x^2 y = \frac{x^4}{4} + C$$

$$y = \frac{x^2}{4} + \frac{C}{x^2}.$$

Solution 7. The integrating factor is $\mu = e^{\int 2\,dx} = e^{2x}$. Multiply through and integrate:

$$\frac{d}{dx}\left[e^{2x} y\right] = 4\,e^{2x}$$

$$e^{2x} y = 2\,e^{2x} + C$$

$$y = 2 + C\,e^{-2x}.$$

Apply the initial condition $y(0) = 1$:

$$1 = 2 + C \implies C = -1$$

$$y = 2 - e^{-2x}.$$

Solution 8. Rewrite as $\frac{dy}{dx} = 1 + \frac{y}{x}$ and substitute $y = vx$, so $\frac{dy}{dx} = v + x\frac{dv}{dx}$:

$$v + x\frac{dv}{dx} = 1 + v$$

$$x\frac{dv}{dx} = 1 \implies dv = \frac{1}{x}\,dx$$

$$v = \ln|x| + C.$$

Back-substitute $v = \frac{y}{x}$:

$$y = x\ln|x| + Cx.$$

Solution 9. Substitute $y = vx$. The right side becomes

$$\frac{x^2 + v^2x^2}{x \cdot vx} = \frac{1 + v^2}{v}.$$

With $\frac{dy}{dx} = v + x\frac{dv}{dx}$,

$$v + x\frac{dv}{dx} = \frac{1 + v^2}{v} = \frac{1}{v} + v$$

$$x\frac{dv}{dx} = \frac{1}{v} \implies v\,dv = \frac{1}{x}\,dx$$

$$\frac{v^2}{2} = \ln|x| + C_1 \implies v^2 = 2\ln|x| + C.$$

Back-substitute $v = \frac{y}{x}$:

$$\frac{y^2}{x^2} = 2\ln|x| + C \implies y^2 = x^2\big(2\ln|x| + C\big).$$

Solution 10. Newton's law of cooling with surrounding temperature $T_s = 20$ and initial temperature $T_0 = 100$ gives

$$T(t) = 20 + (100 - 20)\, e^{-kt} = 20 + 80\, e^{-kt}.$$

Use the reading $T(10) = 60$ to find k:

$$60 = 20 + 80\, e^{-10k} \implies 80\, e^{-10k} = 40$$

$$e^{-10k} = \tfrac{1}{2} \implies -10k = -\ln 2 \implies k = \frac{\ln 2}{10}.$$

Hence the temperature function is

$$T(t) = 20 + 80\, e^{-(\ln 2/10)\, t} = 20 + 80\left(\tfrac{1}{2}\right)^{t/10}.$$

To reach $30°$, solve $T(t) = 30$:

$$30 = 20 + 80\, e^{-kt} \implies 80\, e^{-kt} = 10 \implies e^{-kt} = \tfrac{1}{8}$$

$$-kt = -\ln 8 = -3\ln 2 \implies t = \frac{3\ln 2}{k} = \frac{3\ln 2}{\ln 2/10} = 30 \text{ minutes}.$$

Unit 2 Key Takeaway

Every method in this unit reduces to a single move — rewrite the equation into a form you can integrate. Separation of variables splits a factored rate law; the integrating factor $\mu = e^{\int P\,dx}$ collapses a linear left side into one derivative; and the substitution $y = vx$ turns a homogeneous equation separable. Read the structure first, and the technique always follows.